Real Multi-Reactor $\theta_{13}/\Delta m^2$ Inference at Daya Bay with `tpeanuts`

Fast Simulation of Neutrino Oscillations in Matter

MSc Thesis (Trabajo Fin de Máster) — Inference Companion Document

Juan Ramon Diaz Santos
`diazjuan@alumni.uv.es`

Supervisors: Roberto Ruiz de Austri Bazan & Michele Lucente

September 2026

This document accompanies `notebooks/inference/inference4_dayabay.ipynb` and the `tpeanuts.detector.dayabay`/`tpeanuts.inference` code packages: it documents, end to end, the real Daya Bay Collaboration reactor-antineutrino experiment, the official public data release it is fitted against, the detector-response and reactor-flux physics implemented in `tpeanuts`, the inference machinery (likelihoods, fits, likelihood scans), and the resulting real $\theta_{13}/\Delta m^2_{31}$ measurement, compared against Daya Bay's own published values and against NuFIT. Blue boxes (**Theoretical Foundation**) contain the physics; green boxes (**Implementation**) link that physics to actual `tpeanuts` code; orange boxes collect approximations and limitations; terracotta boxes (**Result**) report a concrete numerical outcome of the notebook. Text highlighted in yellow marks key results/equations.

Contents

Chapter 1

Introduction to the Daya Bay Experiment

1.1 Purpose and how to read this document

This document is the inference-side companion to the *Theoretical Foundations of* `tpeanuts` document: where that document develops the generic neutrino-propagation physics implemented in `core/` and `medium/`, this one documents a single, complete, real-data application built on top of it – fitting the reactor mixing angle θ_{13} and the mass-squared splitting Δm^2_{31} to the real public data release of the Daya Bay reactor antineutrino experiment. Every number quoted here that is not explicitly attributed to a fitted `tpeanuts` result is a real, published, externally verifiable quantity (a detector specification, a reactor parameter, or a Daya Bay Collaboration measurement), traceable through the References chapter.

The five chapters are organised as follows: this chapter introduces the experiment itself – geometry, detector design, and detection principle – with no reference yet to `tpeanuts`; Chapter 2 catalogues the real official data release the code consumes, table by table; Chapter 3 is the physics-and-mathematics core, developing every formula `tpeanuts.detector.dayabay` implements; Chapter 4 documents the statistical inference machinery (likelihoods, fits, likelihood scans) and the three fit strategies used; Chapter 5 presents, interprets, and summarises the numerical results obtained from `inference4_dayabay.ipynb`, compared against Daya Bay's own published values and against NuFIT.

1.2 Overview

The Daya Bay Reactor Neutrino Experiment is a multi-detector, multi-reactor survey of electron-antineutrino disappearance, located at the Daya Bay nuclear power complex in Shenzhen, Guangdong province, China. It was designed with a single, focused physics goal: to measure the smallest of the three PMNS mixing angles, θ_{13}, via the disappearance channel

$$\bar{\nu}_e \to \bar{\nu}_e, \qquad P_{\bar{\nu}_e\bar{\nu}_e}(E, L) \simeq 1 - \sin^2 2\theta_{13} \sin^2\left(\frac{\Delta m^2_{ee} L}{4E}\right) + \left(\theta_{12},\, \Delta m^2_{21} \text{ terms}\right), \tag{1.1}$$

using only reactor-produced $\bar{\nu}_e$ (no beam, no accelerator) and pairs of identically designed detectors at different baselines. Data taking began in December 2011 and continued through 2020, accumulating more than 5.55 million inverse-beta-decay (IBD) candidates with neutron capture on gadolinium (nGd) over 3158 days of operation [2]. The experiment reported the first $> 5\sigma$ observation of a non-zero θ_{13} in March 2012 [1], and subsequently refined the measurement to few-percent precision on both $\sin^2 2\theta_{13}$ and the effective splitting Δm^2_{ee} [3]. This document, and the `tpeanuts` code it accompanies, is built against the Collaboration's own full public data release [5], the same underlying dataset behind [3].

Approximations and Limitations

Scope of this document. `tpeanuts` reproduces the real 6-reactor/8-detector geometry, the real reactor antineutrino flux (with its real non-equilibrium and spent-fuel corrections), the real order-1/M inverse-beta-decay cross section, and the real detector energy response

(IAV, LSNL, Gaussian resolution) at their nominal/central values. It does *not* reproduce the full correlated systematic covariance matrix of the official Daya Bay combined analysis, nor the 6AD/7AD sub-periods (only the 8AD period, all 8 detectors online, is modelled). Chapter 5 reports, honestly and without adjustment, where this simplification does and does not matter for the final $\theta_{13}/\Delta m^2_{31}$ result.

1.3 The reactor complex: six real cores, three nuclear power plants

Theoretical Foundation

The antineutrino sources are six pressurised-water reactor cores, grouped in pairs at three nuclear power plants along the Daya Bay/Ling Ao site [2, 5]:

- **Daya Bay NPP**: reactor cores D1, D2.
- **Ling Ao NPP**: reactor cores L1, L2.
- **Ling Ao-II NPP**: reactor cores L3, L4.

In the official data release and throughout this document (and the `tpeanuts` code), the six cores are labelled generically R1–R6. Each core has a real nominal thermal power of

$$P_{\mathrm{th}} \approx 2.895\ \mathrm{GW_{th}}, \tag{1.2}$$

a single shared, time-averaged value in the data release used identically for all 6 cores (the real per-core power history varies week to week; see Section 3.1). Antineutrinos are produced by the beta decays of neutron-rich fission fragments of four fissile isotopes present in the reactor fuel, $^{235}\mathrm{U}$, $^{238}\mathrm{U}$, $^{239}\mathrm{Pu}$, $^{241}\mathrm{Pu}$, with real time-averaged fission fractions (shared by all 6 cores in this data tier)

$$f_{^{235}\mathrm{U}} \approx 0.563, \quad f_{^{238}\mathrm{U}} \approx 0.076, \quad f_{^{239}\mathrm{Pu}} \approx 0.305, \quad f_{^{241}\mathrm{Pu}} \approx 0.056. \tag{1.3}$$

On average, about six antineutrinos are released per fission, but only those above the inverse-beta-decay threshold ($E_\nu \gtrsim 1.8$ MeV, Chapter 3) are detectable – roughly two per fission, integrated over the real tabulated Huber-Mueller spectra (Chapter 2).

Approximations and Limitations

Every one of the 6 real cores is treated as a point source located at its own real, published position; the finite physical extent of a reactor core introduces a negligible baseline correction (a few centimetres against baselines of hundreds of metres) and is not modelled separately – this matches the official analysis's own documented treatment [2].

1.4 The detector array: three experimental halls, eight antineutrino detectors

Theoretical Foundation

Eight functionally identical antineutrino detectors (ADs) are deployed underground in three experimental halls (EH), chosen to sit at baselines spanning the full oscillatory pattern of (1.1):

- **EH1** (near hall): 2 ADs (AD11, AD12), real flux-weighted baseline ≈ 560 m.
- **EH2** (near hall): 2 ADs (AD21, AD22), real flux-weighted baseline ≈ 600 m.
- **EH3** (far hall): 4 ADs (AD31, AD32, AD33, AD34), real flux-weighted baseline ≈ 1640 m.

"Flux-weighted baseline" means the effective distance obtained by weighting each of the 6 real (detector, reactor) baselines by that reactor's real relative antineutrino output – since every detector actually sees a superposition of all 6 cores, not a single one (Eq. (3.3) in Chapter 3). EH1 and EH2 together form the "near" halls (only weakly oscillated, used as the reference spectrum in a near/far comparison, Chapter 4); EH3 is the "far" hall, sitting close to the first oscillation minimum for the bulk of the reactor antineutrino energy spectrum and hence carrying most of the real θ_{13}-driven deficit.
The detector-naming convention used throughout this document and the `tpeanuts` code, `AD`hd, encodes the hall number h and a sequential detector index d within that hall directly in the name (e.g. `AD31` is the first AD installed at EH3) – this is also how `tpeanuts.detector.dayabay.parameters.NEAR_DETECTORS`/ `FAR_DETECTORS` group detectors by hall (Chapter 3).

Detector construction

Theoretical Foundation

Each AD is built from three nested cylindrical vessels, designed to be as close to functionally identical across all 8 detectors as physically achievable (a deliberate design choice: systematic detector-to-detector differences would otherwise be indistinguishable from a real oscillation effect in a near/far comparison) [2]:

1. **Inner Acrylic Vessel (IAV)**, wall thickness 11 mm, containing 0.1% gadolinium-doped liquid scintillator (GdLS) – the actual neutrino target volume. Gadolinium's huge thermal-neutron capture cross section ($\sigma \sim 4.9 \times 10^4$ b for natural Gd, versus ~ 0.3 b for hydrogen) is what makes the delayed-coincidence tag efficient and fast (Section 1.4).
2. **Outer Acrylic Vessel (OAV)**, filled with undoped liquid scintillator (LS) – a "gamma catcher" that increases the detection efficiency for gamma rays (from both the prompt positron-annihilation signal and the delayed neutron-capture signal) that escape the smaller IAV volume without being fully absorbed there.
3. **Outermost stainless-steel tank**, filled with mineral oil, hosting a total of 192 8-inch photomultiplier tubes (PMTs) radially positioned around the two inner vessels, plus specular reflectors above and below the outer acrylic vessel to improve light collection uniformity.

Three automated calibration units (ACUs), each capable of deploying radioactive sources along one of three vertical axes, sit atop each AD's outer tank, providing the real energy-scale/position calibration data underlying the LSNL and IAV response corrections (Section 3.3).
Each AD is submerged in a two-zone water-Cherenkov muon-veto system (inner and outer water shield, IWS/OWS), instrumented with its own PMTs, providing the real muon-veto efficiency folded into the real daily `eff_livetime` exposure (Section 3.4).

The inverse-beta-decay detection principle

Theoretical Foundation

Every real event this experiment (and this document's inference) is built from is an inverse-beta-decay (IBD) interaction,

$$\bar{\nu}_e + p \to e^+ + n, \tag{1.4}$$

occurring on a free (unbound hydrogen) proton in the GdLS target. IBD is selected via a real, two-part delayed-coincidence signature, occurring within the same detector:

Prompt signal. The positron deposits its kinetic energy locally and then annihilates with an atomic electron, $e^+ + e^- \to 2\gamma$ (2×511 keV), giving a real visible ("prompt") energy

$$E_{\text{prompt}} \approx T_{e^+} + 2m_e = E_\nu - \Delta_{np} + m_e, \qquad \Delta_{np} = M_n - M_p, \tag{1.5}$$

at leading order (the full order-1/M kinematics are developed in Chapter 3).

Delayed signal. The IBD neutron thermalises (over microseconds) and is captured, predominantly on a real gadolinium nucleus (chosen for this purpose precisely because of its large capture cross section), releasing a real ~ 8 MeV gamma-ray cascade ~ 30 μs later on average – well separated in both time and energy from the prompt signal and from most backgrounds.

Selection cuts. A sequence of real cuts (flasher rejection, capture-time window, prompt/delayed energy windows, muon veto, multiplicity) isolates the real IBD candidate sample; Chapter 5 reports how tpeanuts' own real IBD-selection efficiency compares to the official one.

This delayed-coincidence tag is what makes IBD, among the handful of reactions a reactor antineutrino can undergo in a scintillator detector, overwhelmingly the cleanest and most background-suppressible channel – exactly why every reactor-antineutrino oscillation experiment to date (Daya Bay, RENO, Double Chooz, KamLAND) is built around it.

Approximations and Limitations

The real coincidence window, multiplicity cuts, and calibration procedures that turn raw PMT waveforms into a validated IBD candidate list are themselves a substantial piece of the Collaboration's own analysis infrastructure; `tpeanuts` does not reproduce this reconstruction chain, and instead consumes its real, already-processed output directly (the official released event-count tables, Chapter 2) as its external, unmodified real-data input.

1.5 Physics goals and data-taking timeline

Theoretical Foundation

Real data taking proceeded in three real sub-periods, distinguished by how many of the 8 ADs were physically installed and operating:

6AD period (December 2011 – July 2012): 6 ADs online (2 at EH1, 1 at EH2, 3 at EH3) – the dataset behind the original 2012 discovery [1].

8AD period (October 2012 onward): all 8 ADs online (2 at each near hall, 4 at the far hall) – the largest and most-used real dataset, and the *only* period this document's `tpeanuts` model fits (Chapter 3).

7AD period (a later real sub-interval): AD11 temporarily removed from physical data taking.

The physics goal evolved with the data: the 6AD-period near/far rate comparison established $\theta_{13} \neq 0$ [1]; with the accumulated 8AD statistics, a full spectral-shape analysis (comparing the real reconstructed prompt-energy spectrum bin by bin, not just the total rate) additionally constrains the mass-squared splitting via the energy-dependent position of the oscillation minimum, reported as the effective splitting Δm^2_{ee} [3]. Chapter 4 documents both strategies (rate-only and full-spectral-shape) as implemented in `tpeanuts`, plus a real near/far *ratio* fit that recovers both parameters jointly while remaining robust to the detector-response simplifications made in this project (Chapter 5).

Chapter 2

The Official Data Release: Tables and Publications

2.1 Publications

Key References

- **Data release.** Daya Bay Collaboration, *Full Data Release of the Daya Bay Reactor Neutrino Experiment*, v1.0.0, Zenodo, DOI:10.5281/zenodo.17587229 (2025) [5]. The analysis-dataset (TSV) tier is fetched from the official GitHub mirror `github.com/dayabay-experiment/dayabay-data-official` by `notebooks/external/reactor/dayabay/DayaBay1_generator.ipynb` (`zenodo.org` itself returns HTTP 403 from this project's development sandbox) into `data/detector/dayabay/`, in the tidy `.csv` form `tpeanuts.detector.dayabay` consumes.
- **Main precision measurement.** F. P. An et al. (Daya Bay Collaboration), *Precision Measurement of Reactor Antineutrino Oscillation at Kilometer-Scale Baselines by Daya Bay*, Phys. Rev. Lett. **130**, 161802 (2023) [3]. Publishes the $\sin^2 2\theta_{13}/\Delta m^2_{ee}$ best fit compared against this document's own inference in Chapter 5.
- **Discovery paper.** F. P. An et al. (Daya Bay Collaboration), *Observation of Electron-Antineutrino Disappearance at Daya Bay*, Phys. Rev. Lett. **108**, 171803 (2012) [1]. Source of the original near/far rate-comparison strategy Chapter 4 reproduces.
- **Flux and spectrum measurement.** F. P. An et al. (Daya Bay Collaboration), *Improved measurement of the reactor antineutrino flux and spectrum at Daya Bay*, Chin. Phys. C **41**, 013002 (2017) [2]. Source of an independently published per-detector IBD rate table (its Table 5) used in Chapter 5 as an external cross-check of this project's absolute-normalization audit, and of the Reactor Antineutrino Anomaly measurement discussed in Chapter 3.
- **External comparison: NuFIT.** I. Esteban et al., *The fate of hints: updated global analysis of three-flavor neutrino oscillations*, JHEP **12**, 216 (2024), arXiv:2410.05380 [6] – base analysis publication for NuFIT 6.1 (2025), `www.nu-fit.org`. Source of the fixed solar parameters used throughout Chapter 3, and of the independent NuFIT θ_{13} comparison in Chapter 5.

2.2 Data-release structure and format conventions

Theoretical Foundation

The official release ships four parallel data-format tiers (`npz`, `hdf5`, `tsv.bz2`, `root`), each containing the identical physical content; `tpeanuts` consumes the `tsv.bz2` tier only, since it requires no format-specific reader beyond the standard-library `bz2` module and `pandas`. Three real naming conventions appear in the tables catalogued below:

- `{filename}_{key}.tsv.bz2` – a single array, named by its key (e.g. `detector_iav_matrix_iav_matrix.tsv.bz2`).
- `{filename}.tsv/{key}.tsv.bz2` – a folder of related arrays, one file per key (e.g. per-detector IBD spectra).
- `{filename}.tsv/{filename}_{key}.tsv.bz2` – the same, with the folder name repeated in each file for extra clarity when nested paths are otherwise ambiguous.

Real parameter files (`parameters/*.yaml`) additionally carry a `format` field describing their own uncertainty convention (`value`: point value only; `sigma_absolute`: value $\pm$ absolute 1σ; `sigma_percent`: value $\pm$ percent 1σ) and a `state` field (`fixed` or `variable`, i.e. whether the official combined analysis treats that parameter as a free nuisance).

2.3 Table catalogue

Tables 2.1 to 2.4 list every real table this project fetches and the `tpeanuts` loader that reads it (all in `tpeanuts.detector.dayabay.io`), grouped by category; the physics each table encodes is developed in full in Chapter 3.

Table 2.1: Real geometry and reactor-flux tables.

Cached file	Real contents	Loader
`baselines.csv`	(detector, reactor) baseline in km; real $8 \times 6 = 48$-row geometry matrix.	`load_baselines`
`reactor_parameters.csv`	Real fission fractions, energy released per fission (4 isotopes), nominal thermal power 2.895 GW.	`load_reactor_parameters`
`huber_mueller_spectra.csv`	Real per-isotope antineutrino spectra [7, 8, 10], 50 keV mesh, 1.75–13.0 MeV.	`load_huber_mueller_spect`
`nonequilibrium_correction.csv`	Real per-isotope non-equilibrium flux correction (^{235}U, ^{239}Pu, ^{241}Pu only).	`load_nonequilibrium_corr`
`snf_correction.csv`	Real per-reactor spent-nuclear-fuel flux correction, all 6 cores.	`load_snf_correction`
`neutrino_rate_weekly.csv`	Real weekly-resolved per-reactor antineutrino rate, full 2011–2020 data-taking period.	`load_neutrino_rate_weekl`

Table 2.2: Real detector-response and IBD-cross-section tables.

Cached file	**Real contents**	**Loader**
`n_protons.csv`	Real target free-proton count per detector ($\approx 1.43 \times 10^{30}$ nominal, $\pm$ small per-detector correction).	`load_n_protons`
`eres_parameters.csv`	Real 3-term Gaussian energy-resolution coefficients (a, b, c).	`load_eres_parameters`
`detector_efficiency.txt`	Real flat IBD-selection efficiency, $\varepsilon \approx 0.8025$.	`load_detector_efficiency`
`ibd_constants.csv`	Real IBD cross-section constants: $f = 1.0$, $g = 1.2701$, $f_2 = 3.706$, PhaseSpaceFactor$= 1.71465$.	`load_ibd_constants`
`iav_matrix.csv`	Real 240×240 IAV energy-redistribution matrix, 0.05 MeV bins, 0–12 MeV, columns summing to 1.	`load_iav_matrix`
`lsnl_curve_nominal.csv`	Real LSNL nominal nonlinearity curve $f(E) = E_{\rm reco}/E_{\rm true}$.	`load_lsnl_curve`
`lsnl_curve_pulls.csv`	Real 4 LSNL systematic pull curves $f_k(E)$, $k = 0..3$.	`load_lsnl_curve_pulls`

Approximations and Limitations

Not fetched/reproduced: the 6AD/7AD sub-periods (only 8AD); the LSNL/IAV/background official *uncertainty* parameters as a full correlated covariance (their real nominal/central-value curves and rates *are* used, see Table 2.2); and per-detector/per-hall correlation structure for the background-rate systematics (a single correlated scale per category is used instead, Chapter 4).

Table 2.3: Real observed-data and background tables (8AD period).

Cached file	Real contents	Loader
ibd_spectra/ibd_spectrum_{AD}.csv	Real observed IBD prompt-energy candidate spectra, 0.05 MeV bins, 0–12 MeV, one file per detector.	load_ib
background_rates_8AD.csv	Real background rates + uncertainties, 5 categories × 8 detectors.	load_ba
background_shapes/spectrum_shape_{cat}_{AD}.csv	Real normalised background spectral shapes (integrate to 1), same binning as the IBD spectra.	load_ba
exposure_8AD.csv	Real per-detector effective livetime (livetime×eff), summed over 8AD-period days.	load_ex
final_erec_bin_edges.csv	Real (non-uniform) analysis binning, 0.7–12.0 MeV, 27 edges.	load_fi

Table 2.4: Real published-truth and normalization tables.

Cached file	Real contents	Loader
survival_probability_truth.csv	Daya Bay's own real published best fit ($\sin^2 2\theta_{13}$, Δm^2_{32}) and solar input ($\sin^2 2\theta_{12}$, Δm^2_{21}), for comparison only.	load_survival_probabili
global_normalization.txt	Daya Bay's own real nominal free-normalization value (1.0).	load_global_normalizati

Chapter 3

Detector Implementation in `tpeanuts`

This chapter develops, in full mathematical detail, every model `tpeanuts.detector.dayabay` implements: the real multi-reactor antineutrino flux (Section 3.1), the real order-1/M inverse-beta-decay cross section (Section 3.2), the real detector energy response (Section 3.3), efficiency and exposure (Section 3.4), real backgrounds (Section 3.5), and the final composed event-rate assembly (Section 3.6) with its free-normalization and nuisance parameters (Section 3.7).

3.1 Reactor antineutrino flux

`detector/dayabay/flux.py`

Real per-isotope Huber-Mueller spectrum, with real non-equilibrium and spent-nuclear-fuel corrections, a real per-reactor relative power weight, and geometric dilution.

Theoretical Foundation

Per-isotope spectrum and fission-fraction sum

The antineutrino spectrum shape at reactor core r is a real fission-fraction-weighted sum over the four fissile isotopes, with the real non-equilibrium correction $C_i^{\mathrm{neq}}(E)$ folded into each isotope's tabulated spectrum $S_i(E)$ *before* the sum (published for $^{235}\mathrm{U}/^{239}\mathrm{Pu}/^{241}\mathrm{Pu}$ only – no correction exists for $^{238}\mathrm{U}$, which is left uncorrected):

$$\Sigma(E) = \sum_i f_i \, S_i(E) \left(1 + C_i^{\mathrm{neq}}(E)\right). \tag{3.1}$$

The non-equilibrium correction accounts for long-lived fission-daughter beta decays that have not yet reached secular equilibrium with their production rate; it is largest ($\sim 6\%$) just above the IBD threshold and falls to zero by ~ 4.5 MeV.

Absolute normalisation and real per-reactor corrections

The absolute flux at reactor core r, before geometric dilution, combines the real thermal power, the real fission-fraction-weighted mean energy per fission $\bar{e} = \sum_i f_i \bar{e}_i$, a real per-reactor relative power/output weight w_r, and a real per-reactor spent-nuclear-fuel (SNF) correction $C_r^{\mathrm{SNF}}(E)$ (residual antineutrino emission from stored, previously used fuel assemblies near the core):

$$\Phi_{\mathrm{face},r}(E) = \frac{P_{\mathrm{th}}}{\bar{e}} \, w_r \left(1 + C_r^{\mathrm{SNF}}(E)\right) \Sigma(E). \tag{3.2}$$

w_r is derived (not itself a tabulated quantity) from the real weekly antineutrino-rate history, Section 3.1 below.

Geometric dilution and the multi-reactor sum

The flux incident on detector d sums the (independently propagated) contribution of all 6 real cores, each diluted by the real inverse-square baseline law and multiplied by that reactor's own real oscillation survival probability:

$$\Phi_d(E) = \sum_{r=1}^{6} \frac{\Phi_{\mathrm{face},r}(E)}{4\pi L_{r,d}^2}\, P_{\bar\nu_e\bar\nu_e}(E, L_{r,d}). \tag{3.3}$$

$P_{\bar\nu_e\bar\nu_e}(E, L)$ is the standard vacuum 3-flavour survival probability, computed once per (reactor, detector) pair at that pair's own real baseline (Chapter 4); no matter effect is applied (reactor antineutrinos at these energies and baselines cross negligible Earth matter). Equation (3.3) is evaluated independently for every one of the 8 real detectors – each sees its own real 6-baseline combination.

Real per-reactor relative weight

Theoretical Foundation

The official weekly `neutrino_rate` table (Chapter 2) gives each real reactor core's own antineutrino emission rate, resolved week by week across the full 2011–2020 data-taking period, distinguishing the 6AD/7AD/8AD sub-periods via an `n_det` column. Restricting to the real 8AD-period weeks and averaging,

$$w_r = \frac{\langle \mathrm{neutrino\_rate}_r \rangle_{8\mathrm{AD}}}{\frac{1}{6}\sum_{r'} \langle \mathrm{neutrino\_rate}_{r'} \rangle_{8\mathrm{AD}}}, \tag{3.4}$$

a 6-reactor average of exactly 1 by construction. This real per-reactor weight replaces the previous implicit assumption that all 6 cores contribute identically to (3.2) – the real measured weights differ from 1 only at the $\sim$ 1–2% level (real Table 5.1 in Chapter 5), a small but genuine, real per-reactor correction.

Implementation in tpeanuts

- `reactor_spectrum_shape(E_grid_MeV)` implements (3.1) via a cached, corrected copy of the raw Huber-Mueller curves (`_corrected_hm_curves`, built once with the real non-equilibrium correction interpolated onto each isotope's own tabulated energy grid, 0 outside its real defined range).
- `reactor_flux_at_face(E_grid_MeV, reactor)` implements (3.2), with the real SNF correction interpolated onto `E_grid_MeV` (0 outside its own real range).
- `flux_at_detector(E_grid_MeV, baseline_km, reactor)` divides by $4\pi L^2$ (converting `baseline_km` to cm) – one real (reactor, detector) term of (3.3); `detector.dayabay.event_rate.ibd_event_rate` (Section 3.6) performs the sum over the 6 real reactors explicitly.
- `detector.dayabay.parameters.REACTOR_RELATIVE_WEIGHT` implements (3.4) once, at import time, from the real cached weekly-rate table.

3.2 The real order-1/M inverse-beta-decay cross section

`detector/interaction/inverse_beta_decay.py`

Vogel & Beacom's threshold-respecting, order-1/M differential IBD cross section [13], built by numerical $\cos\theta$ *integration using Daya Bay's own real published coupling constants.*

Theoretical Foundation

Zeroth order (no-recoil) baseline

At zeroth order in $1/M$ (M the average nucleon mass), the positron energy is fixed by E_ν alone,

$$E_e^{(0)} = E_\nu - \Delta, \qquad \Delta = M_n - M_p, \tag{3.5}$$

and the total cross section is

$$\sigma^{(0)}(E_\nu) = \sigma_0\,(f^2 + 3g^2)\,E_e^{(0)} p_e^{(0)}, \qquad p_e^{(0)} = \sqrt{\left(E_e^{(0)}\right)^2 - m_e^2}. \tag{3.6}$$

This is Vogel & Beacom's own Eq. (10)-(11), and is the formula this project used for every detector prior to this Daya Bay extension (`sigma_ibd`); it neglects recoil, weak magnetism, and the resulting positron-angle dependence, and its reaction threshold, $E_\nu > ((M_n + m_e)^2 - M_p^2)/(2M_p) \approx 1.806$ MeV, is exact only at this order.

Order-1/M differential cross section

Vogel & Beacom's own Eq. (18)-(19) "high-energy limit" formula explicitly neglects the threshold, an effect their own paper states is "large" below ~ 30 MeV – *every* reactor antineutrino is below 10 MeV, so that formula is not used here. Instead, the full order-1/M formula (their Eq. 13-15), valid down to threshold, is implemented directly. At first order the positron energy depends on the positron-antineutrino scattering angle θ (lab frame):

$$E_e^{(1)} = E_e^{(0)}\left[1 - \frac{E_\nu}{M}(1 - v_e^{(0)}\cos\theta)\right] - \frac{y^2}{M}, \qquad y^2 = \frac{\Delta^2 - m_e^2}{2}, \qquad M = \frac{M_n + M_p}{2}, \tag{3.7}$$

with $v_e^{(0)} = p_e^{(0)}/E_e^{(0)}$, and $p_e^{(1)} = \sqrt{(E_e^{(1)})^2 - m_e^2}$ (zero if the argument is negative, the exact real threshold cutoff at this order). The differential cross section becomes

$$\left(\frac{d\sigma}{d\cos\theta}\right)^{(1)} = \frac{\sigma_0}{2}\left[(f^2 + 3g^2) + (f^2 - g^2)\,v_e^{(1)}\cos\theta\right]E_e^{(1)}p_e^{(1)} - \frac{\sigma_0}{2}\frac{\Gamma}{M}E_e^{(0)}p_e^{(0)}, \tag{3.8}$$

with Γ the weak-magnetism/recoil correction term,

$$\begin{aligned}\Gamma = {} & 2(f + f_2)g\left[(2E_e^{(0)} + \Delta)(1 - v_e^{(0)}\cos\theta) - \tfrac{m_e^2}{E_e^{(0)}}\right] \\ & + (f^2 + g^2)\left[\Delta(1 + v_e^{(0)}\cos\theta) + \tfrac{m_e^2}{E_e^{(0)}}\right] \\ & + (f^2 + 3g^2)\left[(E_e^{(0)} + \Delta)(1 - \tfrac{\cos\theta}{v_e^{(0)}}) - \Delta\right] \\ & + (f^2 - g^2)\left[(E_e^{(0)} + \Delta)(1 - \tfrac{\cos\theta}{v_e^{(0)}}) - \Delta\right]v_e^{(0)}\cos\theta.\end{aligned} \tag{3.9}$$

Removing the apparent threshold singularity

Equation (3.9) contains terms $\propto 1/v_e^{(0)}$, apparently divergent as $E_\nu \to$ threshold ($v_e^{(0)} \to 0$); Vogel & Beacom themselves note this is exactly cancelled by the explicit $p_e^{(0)}$ prefactor multiplying Γ in (3.8). Substituting $v_e^{(0)} = p_e^{(0)}/E_e^{(0)}$ throughout *before* multiplying by

$p_e^{(0)}$ makes this cancellation manifest algebraically, avoiding a numerically risky $0 \times \infty$ evaluation at floating-point precision:

$$
\begin{aligned}
p_e^{(0)}\Gamma = 2(f+f_2)g &\left[p_e^{(0)}(2E_e^{(0)}+\Delta) - \cos\theta\, \frac{(p_e^{(0)})^2}{E_e^{(0)}}(2E_e^{(0)}+\Delta) - \frac{p_e^{(0)} m_e^2}{E_e^{(0)}}\right] \\
&+ (f^2+g^2)\left[\Delta p_e^{(0)} + \cos\theta\,\Delta\,\frac{(p_e^{(0)})^2}{E_e^{(0)}} + \frac{p_e^{(0)} m_e^2}{E_e^{(0)}}\right] \\
&+ (f^2+3g^2)\Big[p_e^{(0)}E_e^{(0)} - \cos\theta\, E_e^{(0)}(E_e^{(0)}+\Delta)\Big] \\
&+ (f^2-g^2)\, p_e^{(0)}\cos\theta\Big[p_e^{(0)} - \cos\theta(E_e^{(0)}+\Delta)\Big],
\end{aligned}
\tag{3.10}
$$

free of any $1/v_e^{(0)}$ or $1/p_e^{(0)}$ term – every division is by $E_e^{(0)}$, which is bounded below by $m_e > 0$ over the whole real IBD kinematic range.

Bare normalisation σ_0

The bare σ_0 (Vogel & Beacom's Eq. 9) does not depend on $f/g/f_2$; it is computed here directly rather than back-derived from the zeroth-order coefficient (which was calibrated using Vogel & Beacom's own, slightly older $g = 1.26$ value, not Daya Bay's real published $g = 1.2701$):

$$
\boxed{\sigma_0 = \frac{G_F^2 \cos^2\theta_C}{\pi}(1+\Delta_{\text{inner}}^R) \times (\hbar c)^2,}
\tag{3.11}
$$

with $\cos\theta_C = 0.974$ the Cabibbo angle and $\Delta_{\text{inner}}^R \approx 0.024$ the energy-independent inner radiative correction, both from [13], and $(\hbar c)^2$ converting from natural units to cm^2/MeV^4. This reproduces Vogel & Beacom's own quoted 0.0952×10^{-42} cm^2/MeV^2 coefficient to $\sim 1\%$ when recombined with their own $f = 1$, $g = 1.26$ – the residual difference is input-value vintage (rounding of $\Delta_{\text{inner}}^R/\cos\theta_C$), not a formula error.

Total cross section and the prompt-energy differential

The total cross section is obtained by numerical integration over $\cos\theta$,

$$
\sigma_{\text{tot}}(E_\nu) = \int_{-1}^{1} d\cos\theta \left(\frac{d\sigma}{d\cos\theta}\right)^{(1)},
\tag{3.12}
$$

the same method Vogel & Beacom themselves used for their own published curves. To build the differential cross section in the observable prompt energy $d\sigma/dT$, each $\cos\theta$-quadrature node's cross-section contribution is redistributed onto the prompt-energy grid at its own order-1/M prompt energy,

$$
E_{\text{prompt}} = E_e^{(1)} + m_e,
\tag{3.13}
$$

via linear-interpolation mass conservation (identical technique to (3.16) below), then converted from a per-quadrature-node mass to a density by dividing by the prompt-energy grid spacing.

Implementation in tpeanuts

- `sigma_ibd_diff_cos_precise(E_nu_MeV, cos_theta, f=1.0, g=1.2701, f2=3.706)` implements (3.7), (3.8) and (3.10) directly, defaulting to Daya Bay's own real coupling constants (`ibd_constants.csv`, Chapter 2); returns $(d\sigma/d\cos\theta,\, E_e^{(1)})$.
- `ibd_cross_section_grid_precise(E_nu_grid_MeV, T_grid_MeV, n_cos=41)` performs the $\cos\theta$-quadrature-and-scatter construction above on a uniform 41-point trapezoidal grid, returning $d\sigma/dT$ on the outer-product grid `(E_nu_grid_MeV, T_grid_MeV)`, shape `(n_E,n_T)`, $\mathrm{cm}^2/\mathrm{MeV}$.
- `_sigma0_bare_cm2_per_mev4()` implements (3.11), cached at module import.
- `sigma_ibd(E_nu_MeV)/ibd_cross_section_grid(...)`: the original zeroth-order formula, (3.6), retained unchanged (used by any detector that has not opted into the precise cross section).
- `detector.common.response.scatter_add_linear(grid, positions, values)`: the shared mass-conserving linear-interpolation scatter utility used both here (redistributing cross-section mass across $\cos\theta$ nodes) and by the LSNL warp matrix (Section 3.3).

Approximations and Limitations

The real, measured effect of this upgrade: the order-1/M cross section is a few percent *below* the zeroth-order one, growing with E_ν (consistent with Vogel & Beacom's own stated $\mathcal{O}(E_\nu/M)$ scaling, with an $\mathcal{O}(-7)$ numerical coefficient) – validated independently in this project's own test suite (`detector/dayabay/test/test2_response_and_fit.py`).

3.3 Detector energy response: IAV, LSNL, and Gaussian resolution

`detector/dayabay/response.py`

The full real response chain, applied in the official Daya Bay Collaboration analysis pipeline's own order: IAV redistribution, then LSNL nonlinearity, then Gaussian photostatistics resolution.

Theoretical Foundation

Real response order

The official Daya Bay dagflow/GNA analysis pipeline's own node names (confirmed directly from the Collaboration's own CHEP 2026 conference presentation on that framework) read, cumulatively,

$$\text{raw} \longrightarrow \text{stages.iav} \longrightarrow \text{stages.evis (IAV+LSNL)} \longrightarrow \text{stages.erec (+resolution)}, \tag{3.14}$$

i.e. IAV is applied to the true deposited energy *before* LSNL, which is then followed by the Gaussian resolution smearing. This also matches the physical picture: energy lost into the inert acrylic vessel material near its wall (the IAV effect) never produces scintillation light, so it cannot be subject to a light-yield nonlinearity curve (LSNL) –

only the light-producing, IAV-redistributed energy is.

IAV redistribution

The real IAV correction is a 240×240 energy-redistribution matrix R^{IAV} on a 0.05 MeV, 0–12 MeV grid (matching this project's own `T_GRID_MEV` exactly), normalised so that each column sums to 1 – a genuine discrete probability-mass (not density) transfer matrix,

$$R_{ij}^{\mathrm{IAV}} = P(\text{reconstructed in bin } i \mid \text{true energy in bin } j), \qquad \sum_i R_{ij}^{\mathrm{IAV}} = 1. \tag{3.15}$$

LSNL nonlinearity

The real LSNL correction is a multiplicative energy-scale warp, $f(E) = E_{\text{reconstructed}}/E_{\text{true deposited}}$, mapping true energy T to $T_{\mathrm{nl}} = T \cdot f(T)$. Represented as a mass-conserving matrix on the same grid, column j (true energy T_j) is split by linear interpolation between the two grid points bracketing $T_{\mathrm{nl}}(T_j)$:

$$\boxed{R_{ij}^{\mathrm{LSNL}} = \begin{cases} 1 - w_j, & i = \lfloor T_{\mathrm{nl}}(T_j) \rfloor_{\mathrm{grid}}, \\ w_j, & i = \lceil T_{\mathrm{nl}}(T_j) \rceil_{\mathrm{grid}}, \\ 0, & \text{otherwise}, \end{cases}} \qquad w_j = \frac{T_{\mathrm{nl}}(T_j) - T_{\lfloor \cdot \rfloor}}{T_{\lceil \cdot \rceil} - T_{\lfloor \cdot \rfloor}} \tag{3.16}$$

(clamped to $[0, 1]$, with all mass dumped at the nearest edge for a position outside the grid range) – the identical construction used for the IBD cross section's $\cos\theta \to T$ redistribution (Section 3.2). By construction $\sum_i R_{ij}^{\mathrm{LSNL}} = 1$ for every column j.

Official LSNL pull-curve model

The real LSNL curve used is not fixed to its nominal shape alone: the official systematic model is a linear combination of the nominal curve f_0 and 4 real published pull curves f_k,

$$\boxed{f(E) = f_0(E) + \sum_{k=1}^{4} a_k \big(f_k(E) - f_0(E)\big),} \tag{3.17}$$

with each a_k a real free nuisance carrying an official independent, zero-mean, unit-sigma Gaussian prior (`parameters/detector_lsnl.yaml`). At $a_k = 0$ for all k, (3.17) reduces to the nominal curve; Chapter 4 documents how a_k enters the fit as a real, Gaussian-constrained nuisance parameter.

Gaussian photostatistics resolution

The real 3-term resolution formula,

$$\boxed{\frac{\sigma(E)}{E} = \sqrt{a^2 + \frac{b^2}{E} + \frac{c^2}{E^2}}}, \qquad a = 0.016,\ b = 0.081\ \sqrt{\mathrm{MeV}},\ c = 0.026\ \mathrm{MeV}, \tag{3.18}$$

with terms interpretable as spatial/temporal non-uniformity (a), photon (photoelectron) counting statistics (b), and PMT dark noise/electronics (c), builds a genuine probability *density* in T' (not a discrete mass matrix, unlike IAV/LSNL): $R^{\mathrm{Gauss}}(T'|T) = \mathcal{N}(T'; T, \sigma(T))$.

Composition

All three steps are real, independent, official detector-response ingredients (the data release ships them as separate real files precisely because they are physically distinct effects); they are composed as sequential linear operators. Because the folding machinery (Section 3.6) integrates $R(T'|T)$ against T via trapezoidal quadrature, every step is first built as a discrete, mass-conserving transfer matrix on the shared 0.05 MeV grid, and only the final *composed* matrix is converted from probability mass to the density the folding integral expects (dividing by the grid spacing δT):

$$\boxed{R(T'|T) = \frac{1}{\delta T}\, R^{\mathrm{Gauss}}_{\mathrm{mass}}\; R^{\mathrm{LSNL}}\; R^{\mathrm{IAV}},} \tag{3.19}$$

where $R^{\mathrm{Gauss}}_{\mathrm{mass}} = \delta T \cdot R^{\mathrm{Gauss}}$ (density converted to an approximate mass for the composition, valid since $\delta T = 0.05$ MeV is fine compared to $\sigma(E)$ over the whole relevant energy range). Reading (3.19) as a matrix product applied to a column vector v representing a true-energy spectrum, $R(T'|T)\, v = R^{\mathrm{Gauss}}_{\mathrm{mass}}(R^{\mathrm{LSNL}}(R^{\mathrm{IAV}} v))/\delta T$ – v is transformed by R^{IAV} first, then R^{LSNL}, then $R^{\mathrm{Gauss}}_{\mathrm{mass}}$, exactly the real physical order stated above.

Implementation in tpeanuts

- `sigma_MeV(T_grid_MeV, a, b, c)` implements (3.18).
- `_iav_mass_matrix(T_grid_MeV)` embeds the real 240×240 `IAV_MATRIX` into the project's own (possibly larger) grid, with an identity fallback for any extra grid point beyond the real IAV table's own range (physically negligible: the real IBD prompt-energy spectrum is essentially zero there).
- `_lsnl_curves_on_grid(T_grid_MeV)` interpolates the real nominal/pull curves onto the shared grid once (`@torch.no_grad()`, fixed real data).
- `_lsnl_warp_matrix(T_grid_MeV, lsnl_pulls=None)` implements (3.16) and (3.17); differentiable w.r.t. `lsnl_pulls` (only the linear combination and the resulting warp weights carry gradient – the underlying curve interpolation is fixed real data).
- `response_matrix(T_grid_MeV, Tprime_grid_MeV, a, b, c, lsnl_pulls=None)` implements (3.19).

Approximations and Limitations

Every component above uses the real official nominal/central-value curves/matrix; the LSNL pull-curve nuisances a_k can be fit (Chapter 4), but the IAV matrix's own official off-diagonal-scale systematic (`detector_iav_offdiag_scale.yaml`, a single real nuisance with a 4% prior) is not implemented. The three response steps are composed assuming they act as independent, sequential linear operators – a standard simplification when a full joint response covariance is not being reproduced.

3.4 Efficiency and exposure

Theoretical Foundation

The real IBD-selection efficiency $\varepsilon \approx 0.8025$ (Gd-capture fraction $\times$ analysis-cut efficiencies, consistent with the real published individual cut efficiencies – Gd-capture fraction $\approx 85.5\%$, prompt-energy cut $\approx 99.81\%$, capture-time cut $\approx 98.70\%$, multiplicity cut $\approx 97.5\%$, flasher cut $\approx 99.98\%$, whose product is $\approx 81.6\%$, close to the quoted 80.25%) is applied as a flat multiplicative factor, on top of and separate from the real per-detector, per-day effective livetime (`eff_livetime` = `livetime`×`eff`, the latter itself the real combined muon-veto and multiplicity-cut daily efficiency), summed over the real 8AD-period days to give each detector's total real exposure in seconds.

Implementation in tpeanuts

`DETECTOR_EFFICIENCY` (a flat scalar) multiplies the folded reconstructed spectrum after response and before binning (Section 3.6); `load_exposure()` sums `eff_livetime` over real 8AD-period days per detector.

3.5 Real backgrounds

Theoretical Foundation

Five real background categories contribute to the observed spectrum, each with its own real published per-detector rate (events/day) and real normalised shape histogram (integrating to 1 on the same 0.05 MeV binning as the signal):

- **Accidentals** – uncorrelated prompt-like/delayed-like triggers randomly coincident in time; the real best-measured category (fractional rate uncertainty $\sim 0.1\%$).
- ^{9}**Li**/^{8}**He** – cosmogenic isotopes produced by through-going muons, beta-decaying with a delayed-neutron-emission branch that mimics the IBD signature.
- **Fast neutrons** – energetic neutrons from muon spallation outside the water shield, which can deposit a prompt-like recoil energy followed by a real neutron capture.
- **AmC** – correlated background from the AmC neutron calibration sources themselves.
- $^{13}\mathbf{C}(\alpha, n)^{16}\mathbf{O}$ – alpha decays of natural radioactivity in the scintillator inducing a real (α, n) reaction on carbon.

The predicted background count per real analysis bin is

$$N_k^{\text{bkg}} = \sum_{\text{category}} (\text{rate}_{\text{category}} \times \text{exposure\_days}) \times \text{shape}_{\text{category},k}, \tag{3.20}$$

with the fine-binned (0.05 MeV) product rebinned onto the real non-uniform analysis edges by a plain per-bin sum (every real analysis-bin edge lands exactly on a fine-bin boundary, so no fine bin straddles two coarse bins).

Implementation in tpeanuts

`real_background_counts(detector, exposure_seconds, bin_edges_MeV, category_scale=None)` implements the sum above; `rebin_discrete_counts` performs the fine-to-coarse rebinning. An optional `category_scale` dict multiplies each category's contribution by a real per-category nuisance (Section 3.7).

3.6 Composed event-rate assembly

`detector/dayabay/event_rate.py`

The one place this package's real 6-reactor geometry, real flux, real order-1/M cross section, real response, and real background come together, per detector.

Theoretical Foundation

For one real detector d, the predicted signal (pre-response) spectrum folds the multi-reactor flux sum (3.3) with the order-1/M differential cross section (3.8),

$$\frac{dR}{dT}(T) = N_p^{(d)} \int dE_\nu \; \Phi_d(E_\nu)\, \frac{d\sigma}{dT}(E_\nu, T), \tag{3.21}$$

with $N_p^{(d)}$ the real target free-proton count for detector d; the response matrix (3.19) is then applied,

$$\frac{dR}{dT'}(T') = \int dT \; R(T'|T)\, \frac{dR}{dT}(T), \tag{3.22}$$

followed by the flat efficiency factor and integration into the real non-uniform analysis bins,

$$N_k(\theta) = \nu_{\text{norm}} \cdot \text{exposure} \cdot \varepsilon \int_{\text{bin } k} dT' \; \frac{dR}{dT'}(T';\theta) \;+\; N_k^{\text{bkg}}, \tag{3.23}$$

the full, real, per-bin, per-detector prediction, with ν_{norm} the free global-normalization nuisance (Section 3.7) multiplying the signal only (the real background is separately measured and not rescaled by it).

Implementation in tpeanuts

- `ibd_event_rate(detector, p_ee_per_reactor, ..., signal_scale=1.0, background_category_scale=None, lsnl_pulls=None)` implements (3.23) end to end, calling `detector.common.event_rate.predicted_counts` with the real cross section, real response, real target, real exposure, and real background plugged in.
- `p_ee_per_reactor`: `{reactor: P_ee(E)}`, one tensor per real reactor, already oscillated at that reactor's own real baseline – supplied by `detector.dayabay.inference_model.DayaBayDetectorModel` (Chapter 4), which itself calls `tpeanuts.medium.vacuum.oscillation_model.VacuumOscillationModel.predict_pee` once per reactor.

3.7 Free normalization and nuisance parameters

Theoretical Foundation

Global normalization

Daya Bay's own official combined analysis leaves a free overall normalization nuisance in its fit (`parameters/detector_normalization.yaml`, real nominal value 1.0, labelled "Global IBD rate normalization"): even Daya Bay's own simulation chain is not asserted to self-normalize to the observed rate without fitting it. Equation (3.23)'s ν_{norm} is this same real, official parameter, not an ad hoc rescaling invented for this project – see Chapter 5 for why it is needed (a real, documented $\sim 19\%$ absolute-normalization gap, audited and explained in Chapter 5, not attributable to any single unit or loading error).

Real background-rate nuisances

Five further real nuisances, one Gaussian-constrained multiplicative scale c_k per background category (nominal 1), with real prior width

$$\sigma_k = \frac{\sqrt{\sum_d (\text{uncertainty}_{k,d})^2}}{\sum_d \text{rate}_{k,d}}, \tag{3.24}$$

a rate-weighted quadrature combination of the real per-detector (rate, uncertainty) pairs into one correlated scale per category – a simplification of the official per-detector/per-hall correlation structure (accidentals genuinely uncorrelated between detectors; AmC correlated across all detectors; $^{9}\text{Li}/^{8}\text{He}$ and fast neutrons correlated within a hall), not a per-detector free parameter.

Real LSNL pull-curve nuisances

The 4 real a_k of (3.17), official prior independent zero-mean unit-sigma Gaussian.

Implementation in tpeanuts

- `DayaBayDetectorModel(normalization_free, background_free, lsnl_free)`: each flag appends the corresponding real nuisance name(s) to `.free` and slices them from `theta` in `.predict` (Chapter 4).
- `detector.dayabay.parameters.BACKGROUND_CATEGORY_SIGMA` implements (3.24) once, at import time.
- `tpeanuts.inference.likelihood.gaussian_prior_penalty(values, mean, sigma)`: the generic, reusable $-2\ln L$ Gaussian-prior penalty added to the fit's likelihood for any of these nuisances (Chapter 4).

Chapter 4

The Inference Process

This chapter documents the statistical inference machinery: the 3-flavour vacuum oscillation probability every model below shares (Section 4.1), the three composed model variants (Section 4.2), the likelihood statistics (Section 4.3), the gradient-based fit procedure and its uncertainty estimate (Section 4.4), and the 2-D likelihood scan (Section 4.5).

4.1 Real 3-flavour vacuum oscillation probability

`medium/vacuum/oscillation_model.py`

Shared, detector-agnostic 3-flavour vacuum oscillation model (`VacuumOscillationModel`), reused unchanged from the rest of `tpeanuts`.

Theoretical Foundation

Reactor antineutrinos at Daya Bay's real baselines ($\lesssim 2$ km) cross negligible Earth matter, so the exact coherent 3-flavour *vacuum* survival probability is used,

$$P_{\bar\nu_e\bar\nu_e}(E, L; \theta) = \left| \sum_i U_{ei}^* \, e^{-i\Delta m^2_{i1} L/2E} \, U_{ei} \right|^2, \tag{4.1}$$

with U the PMNS matrix [9, 12] and antineutrino sign convention ($U \to U^*$ relative to the neutrino case). Free parameters throughout this document: θ_{13} and $\Delta m^2_{3l} \equiv \Delta m^2_{31}$ (normal ordering); fixed at Daya Bay's own real external solar input, $\theta_{12} = \frac{1}{2} \arcsin \sqrt{\sin^2 2\theta_{12}}$, Δm^2_{21} (`survival_probability_solar.yaml`, Chapter 2). θ_{23}/δ_{CP} do not affect (4.1) (a direct consequence of the standard 3-flavour PMNS parametrisation) and are held at an arbitrary NuFIT preset value [6].

Implementation in tpeanuts

`VacuumOscillationModel.predict_pee(theta, L_km, E_grid_MeV)` implements (4.1) exactly (full 3-flavour, not a 2-flavour approximation), called once per (reactor, detector) pair by `DayaBayDetectorModel.predict` (Section 4.2).

4.2 The three model variants

`detector/dayabay/inference_model.py`

`DayaBayDetectorModel`, `JointDayaBayModel`, `NearFarRatioDayaBayModel`, `DayaBayExperimentalModel`: composition layer wrapping `VacuumOscillationModel` around `event_rate.ibd_event_rate` (Chapter 3).

Per-detector and joint absolute-count models

Theoretical Foundation

`DayaBayDetectorModel` evaluates (4.1) once per real reactor (6 calls, each at that reactor's own real baseline), then calls (3.23) to fold the real 6-reactor flux sum through the real cross section, response, and background. Its free-parameter vector is assembled, in order, from

$$\theta = \underbrace{(\theta_{13},\, \Delta m^2_{3l})}_{\text{oscillation}} \oplus [\nu_{\text{norm}}] \oplus [c_1, \ldots, c_5] \oplus [a_1, \ldots, a_4], \tag{4.2}$$

with each bracketed group present only if the corresponding `normalization_free`/`background_free`/`lsnl_free` flag is set. `JointDayaBayModel` concatenates several `DayaBayDetectorModel` predictions (typically all 8 real detectors) sharing one θ, for a full combined fit – the **full-spectral-shape** strategy.

Rate-only model

Theoretical Foundation

A rate-only observable integrates (3.23) over the full real analysis-bin range, discarding the intra-detector spectral shape but retaining the real detector-to-detector rate deficit:

$$N_d^{\text{rate}}(\theta) = \sum_k N_k^{(d)}(\theta), \tag{4.3}$$

one real number per detector. Because a rate-only observable has essentially no independent Δm^2_{31} leverage in this two-baseline-tier geometry, Δm^2_{31} is fixed at Daya Bay's own published value and only θ_{13} (plus ν_{norm}) is fit.

Near/far ratio model

Theoretical Foundation

Mirroring Daya Bay's own original 2012 discovery strategy [1], the far-hall/near-hall per-bin count ratio is predicted instead of absolute counts:

$$R_k(\theta) = \frac{N_k^{\text{far}}(\theta)}{N_k^{\text{near}}(\theta)}, \qquad N_k^{\text{near,far}}(\theta) = \sum_{d\,\in\,\text{NEAR,FAR}} N_k^{(d)}(\theta), \tag{4.4}$$

with NEAR $= \{\text{AD11}, \text{AD12}, \text{AD21}, \text{AD22}\}$ (EH1+EH2) and FAR $= \{\text{AD31}, \ldots, \text{AD34}\}$ (EH3, Section 1.4). This cancels, by construction, every systematic common to all 8 real detectors: with `normalization_free =False`, ν_{norm} is not even a free parameter of this model (it would cancel in (4.4) to the extent the real background is negligible relative to signal, and only approximately once it is not); any *residual* mismatch in the simplified IAV/LSNL/cross-section response also cancels approximately, since it multiplies the near and far prediction identically.

Experimental reparametrisation

Theoretical Foundation

Daya Bay's own combined analyses report $\sin^2(2\theta_{13})$ and the effective splitting

$$\Delta m^2_{ee} = \cos^2\theta_{12}\,\Delta m^2_{31} + \sin^2\theta_{12}\,\Delta m^2_{32} \tag{4.5}$$

[3, 11], not the bare PMNS $(\theta_{13}, \Delta m^2_{3l})$ this project's models natively expose. With $\theta_{12}/\Delta m^2_{21}$ fixed, substituting $\Delta m^2_{32} = \Delta m^2_{31} - \Delta m^2_{21}$ into (4.5) and solving gives an exact, fixed linear offset,

$$\Delta m^2_{31} = \Delta m^2_{ee} + \sin^2\theta_{12}\,\Delta m^2_{21}, \tag{4.6}$$

so `DayaBayExperimentalModel` is a pure, exactly invertible change of variables (verified in this project's own test suite), not a different fit – it exists purely so a fit's free-parameter vector and any resulting contour are directly comparable to Daya Bay's own published numbers/plots.

Implementation in tpeanuts

- `DayaBayDetectorModel(oscillation_model, detector, normalization_free=True, background_free=False, lsnl_free=False)`: `.free` implements (4.2); `.predict(theta)` slices θ in that same order and calls `ibd_event_rate`.
- `JointDayaBayModel(models)`: `.predict` concatenates.
- `NearFarRatioDayaBayModel(near_models, far_models)` implements (4.4); `.real_observed_ratio()` (a `@staticmethod`) returns the real observed ratio and its propagated uncertainty (Section 4.3).
- `DayaBayExperimentalModel(model, theta12, dm21)` implements (4.6), converting `theta`'s first two entries before delegating to the wrapped model.

4.3 Likelihood statistics

`inference/likelihood.py`

`poisson_nll`, `chi2_asymmetric`, `gaussian_prior_penalty` – every $-2\ln L$ statistic used in this document.

Theoretical Foundation

Poisson (Baker-Cousins) likelihood

Absolute per-bin counts (full-shape and rate-only fits) are compared via the Poisson likelihood-ratio statistic [4],

$$-2\ln L_{\text{Poisson}} = 2\sum_k \Big[N_k - n_k^{\text{obs}} + n_k^{\text{obs}} \ln \frac{n_k^{\text{obs}}}{N_k}\Big], \tag{4.7}$$

with N_k/n_k^{obs} the predicted/observed counts. The extra $-n_k^{\text{obs}} + n_k^{\text{obs}} \ln n_k^{\text{obs}}$ terms subtract the saturated model's own likelihood, so (4.7) is exactly 0 when $N_k = n_k^{\text{obs}}$ bin by bin and behaves like an ordinary chi-square asymptotically (Wilks' theorem).

Gaussian (chi2_asymmetric) likelihood

The near/far ratio (a ratio of counts, not itself Poisson-distributed) is compared via a Gaussian statistic with one-sided uncertainty,

$$-2\ln L_{\chi^2} = \sum_k \left(\frac{R_k - R_k^{\text{obs}}}{\sigma_k} \right)^2, \tag{4.8}$$

with the real observed ratio's uncertainty propagated from independent Poisson numerator/denominator counts,

$$\sigma_k = R_k^{\text{obs}} \sqrt{\frac{1}{N_k^{\text{far,obs}}} + \frac{1}{N_k^{\text{near,obs}}}}. \tag{4.9}$$

Gaussian prior penalty

Real, externally measured nuisance-parameter constraints (background-rate scales, LSNL pulls, Section 3.7) are added as an independent Gaussian penalty on the free-parameter vector itself, not the prediction:

$$-2\ln L_{\text{penalty}} = \sum_j \left(\frac{\theta_j - \mu_j}{\sigma_j} \right)^2, \tag{4.10}$$

added to (4.7) (or (4.8)) to give the total statistic minimized.

Implementation in tpeanuts

- `poisson_nll(prediction, value)` implements (4.7), built as `xlogy(value,value)-xlogy(value,prediction)` rather than the algebraically equal single-`xlogy` form, to keep the backward pass finite at an empty bin (`value==0`).
- `chi2_asymmetric(prediction, value, sigma_minus, sigma_plus)` implements (4.8).
- `gaussian_prior_penalty(values, prior_mean, prior_sigma)` implements (4.10); passed as `fit_lbfgs`'s `penalty_fn` argument (Section 4.4).
- `NearFarRatioDayaBayModel.real_observed_ratio` implements (4.9).

4.4 Gradient-based fit procedure

`inference/fit.py`

`fit_lbfgs/minimize_lbfgs`: LBFGS point estimate plus Laplace/Fisher covariance, for any `DifferentiableModel`.

Theoretical Foundation

Parameter rescaling

LBFGS's identity-initialised inverse-Hessian approximation assumes roughly comparable parameter scales; $\theta_{13} \sim \mathcal{O}(1)$ rad and $\Delta m^2_{31} \sim \mathcal{O}(10^{-3})$ eV^2 differ by 3-4 orders of magnitude, which starves the line search of a usable step size. A dimensionless $u = \theta/\text{scale}$ is fit instead, with scale set from the loss gradient at θ_0,

$$\text{scale} = \left|\nabla_\theta(-2\ln L)\right|^{-1}_{\theta_0}, \tag{4.11}$$

undone only in the returned $\hat{\theta}$ – every model always sees physical-unit θ.

Optimisation and convergence

$\hat{\theta} = \arg\min_\theta[-2\ln L(\theta) + \text{penalty}(\theta)]$ is found by `torch.optim.LBFGS` using the exact autograd gradient of the total statistic through every real physics step of Chapter 3 (no finite differences).

Laplace/Fisher covariance

The marginal uncertainty on each fitted parameter is the Laplace approximation: with $\mathcal{I} = \frac{1}{2}\nabla^2_\theta(-2\ln L)$ the Fisher information at the optimum (the autograd Hessian of the total, penalised statistic),

$$\Sigma = \mathcal{I}^{-1}, \qquad \sigma_j = \sqrt{\Sigma_{jj}}. \tag{4.12}$$

A direction the data barely constrains gives a Hessian eigenvalue near zero (occasionally slightly negative from numerical round-off); small or negative eigenvalues are floored at a tiny fraction of the largest one before inversion, so an unconstrained direction gets a large but finite σ_j instead of `nan`.

Implementation in tpeanuts

- `fit_lbfgs(model, theta0, value, sigma_minus=None, sigma_plus=None, likelihood=..., penalty_fn=None, max_iter=200)` returns a **`FitResult`** (`theta_hat`, `param_names`, `chi2_history`, `covariance`, and a `sigma` property).
- `penalty_fn`: an optional `Callable[[theta], Tensor]` added to the loss inside the LBFGS closure *and* to the Hessian computation for (4.12) – a tight prior correctly shrinks its own parameter's reported uncertainty. `None` (default) reproduces the previous, unconstrained-fit behaviour exactly.

4.5 2-D likelihood scan

`inference/scan.py`

`loglik_grid`: evaluates $-2\ln L$ on a (θ_x, θ_y) grid, for visual/contour diagnostics.

Theoretical Foundation

On a chosen 2-D grid, every entry of `model.free` other than the two scanned parameters is held fixed at a baseline θ (a cheap slice through the likelihood surface) or re-minimised at each grid point (the statistically correct construction when other parameters are also uncertain, at the cost of one LBFGS run per grid point). With exactly two free parameters in `model.free` (the rate-only and near/far models, once their own nuisances are fixed/absent), nothing remains to profile and both modes coincide – the scans in Chapter 5 are genuine 2-D likelihood surfaces of the model's full free set, not fixed-nuisance slices.

Implementation in tpeanuts

`loglik_grid(model, theta_baseline, param_x, x_grid, param_y, y_grid, value, sigma_minus=None, sigma_plus=None, likelihood=..., profile_others=False)` returns a real tensor shaped `(nx,ny)`; subtracting its minimum and comparing against standard 2-D $\Delta\chi^2$ confidence levels draws confidence contours.

Chapter 5

Results

This chapter presents, interprets, and summarises the real numerical results of `notebooks/inference/inferenc` executed end to end against the real Daya Bay 8-detector, 8AD-period data (Chapter 2) using the physics of Chapter 3 and the inference machinery of Chapter 4. Every number below is a real, reproducible output of that notebook, not illustrative.

5.1 Real per-reactor relative weight

Table 5.1: Real per-reactor relative power weight w_r ((3.4)), derived from the real weekly antineutrino-rate history, 8AD period.

Reactor	R1	R2	R3	R4	R5	R6
w_r	1.0010	0.9905	1.0126	0.9965	1.0079	0.9914

Table 5.1 shows the real values obtained: a small but genuine ±1–1.5% real reactor-to-reactor spread around the 6-reactor average of 1, replacing the previous implicit assumption that all 6 real cores contribute identically to the flux sum (Eq. (3.2)).

5.2 Closure at Daya Bay's own published truth

Result

Evaluating the full real 8-detector model at Daya Bay's own published $\theta_{13}/\Delta m^2_{31}$, with $\nu_{\text{norm}} = 1$ (no fit), gives

$$N^{\text{total}}_{\text{pred}} = 3{,}705{,}407, \qquad N^{\text{total}}_{\text{obs}} = 3{,}114{,}934, \qquad \frac{N_{\text{pred}}}{N_{\text{obs}}} = 1.190, \tag{5.1}$$

i.e. the simulation over-predicts the real observed rate by $\approx 19\%$, remarkably uniformly across all 8 detectors (1.179–1.195, Table 5.2). Gradients w.r.t. θ_{13}, Δm^2_{31}, and ν_{norm} are all finite and non-zero, confirming autograd flows correctly through the real multi-reactor flux, the real order-1/M cross section, and the real IAV+LSNL+Gaussian response chain before any fit is attempted.

5.3 Absolute-normalization audit

A dedicated audit of the real $\sim 19\%$ gap above checked every component of the absolute-rate chain: the real target proton count and detector efficiency (both loaded directly from the official release, not derived); the order-1/M cross section (independently validated to $\sim 1\%$ against Vogel & Beacom's own quoted coefficient, Chapter 3); the real Huber-Mueller antineutrino

Table 5.2: Predicted/observed ratio per detector at Daya Bay's own published truth, $\nu_{\text{norm}} = 1$.

Detector	Predicted	Real observed	Ratio
AD11	827,463	695,290	1.190
AD12	840,090	704,771	1.192
AD21	792,729	667,823	1.187
AD22	782,829	658,020	1.190
AD31	116,030	97,069	1.195
AD32	116,150	97,504	1.191
AD33	115,008	96,793	1.188
AD34	115,108	97,664	1.179

yield (the real table spans only 1.75–13.0 MeV, integrating to ≈ 2.0 antineutrinos/fission above the IBD threshold – the correct expected value for that truncated real range, not a bug); and the thermal-power-to-fission-rate conversion (a standard textbook formula). No unit or normalization error was found.

Result

Table 5.3 cross-checks this notebook's real observed and predicted rates-per-day against an *independently published* table (Table 5 of [2], a different, earlier 621-day dataset). The real *observed* rate matches the published table to 1–3% for every detector, confirming the real data/exposure loading is correct; the *predicted* rate exceeds it by the same $\sim$ 19–22% already seen above, confirming the gap lives entirely in the forward-model normalization chain, not in a data-loading or exposure error.

Table 5.3: Real observed/predicted rate per day vs. an independently published rate table [2].

Detector	tpeanuts [1/day]	Real obs. [1/day]	Published [1/day]	obs/pub.	pred/pub.
AD11	796.0	668.9	657.18	1.018	1.211
AD12	810.7	680.1	670.14	1.015	1.210
AD21	723.6	609.6	594.78	1.025	1.217
AD22	714.9	600.9	590.81	1.017	1.210
AD31	90.3	75.6	73.90	1.022	1.222
AD32	90.4	75.9	74.49	1.019	1.214
AD33	89.5	75.4	73.58	1.024	1.217
AD34	89.6	76.0	75.15	1.012	1.193

Part of the real gap is a known, published physical effect: the Huber-Mueller flux model itself is measured to predict $\approx$ 5–6% more than the real observed rate – the "Reactor Antineutrino Anomaly" [2], quoted there as a measured/predicted flux ratio of 0.946 ± 0.020 for the Huber+Mueller model. The remainder is consistent with the many further real correlated systematics (target-mass, efficiency, and flux-normalization nuisances) Daya Bay's own official combined fit also carries and this project does not reproduce individually – precisely why the real official $\nu_{\rm norm}$ nuisance (Section 3.7) exists rather than being fixed to 1.

5.4 Full-spectral-shape fit: three attempts, one honest diagnosis

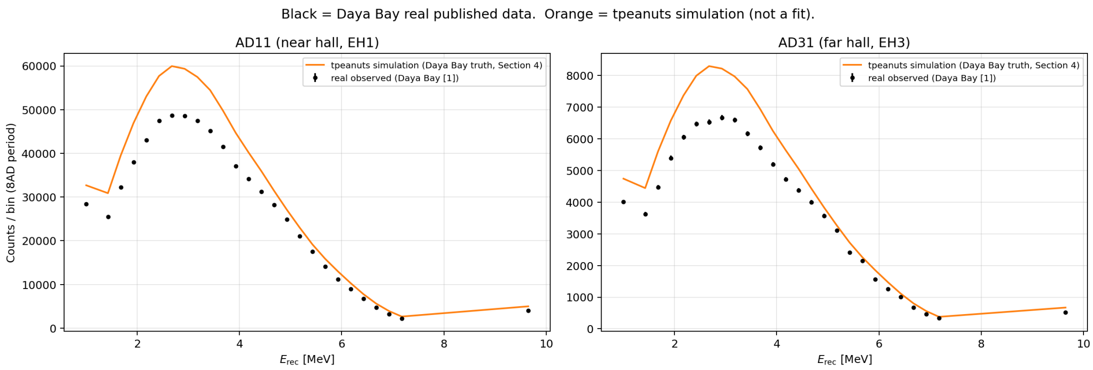

Figure 5.1: Predicted (tpeanuts, at Daya Bay's own published truth) vs. real observed prompt-energy spectra, near-hall (AD11) and far-hall (AD31) detectors.

Figure 5.1 shows the real observed spectrum against the `tpeanuts` simulation at Daya Bay's own published truth for one near-hall and one far-hall detector: both follow the same overall shape (rising then falling with a broad IBD-kinematics-driven peak around 3–4 MeV), with the far-hall spectrum visibly suppressed relative to its no-oscillation expectation, the real qualitative signature this whole measurement is built on.

Minimising (4.7) jointly over $(\theta_{13}, \Delta m^2_{31}, \nu_{\rm norm})$ against the real concatenated 8-detector, 208-bin spectrum was attempted three times, with increasingly rich nuisance-parameter freedom, summarised in Table 5.4.

Table 5.4: Full-spectral-shape fit: three real attempts (Section 5.1-5.3 of the notebook).

Attempt	$\hat\theta_{13}$ [deg]	$\Delta\hat m^2_{31}$ [10^{-3}eV^2]	$\hat\nu_{\rm norm}$	$-2\ln L$ final
Published truth	8.506	2.528	–	–
Unconstrained (5.1)	9.172	7.483	0.874	6445.8
+ Background nuisances (5.2)	9.219	7.497	0.875	6390.6
+ LSNL pull nuisances (5.3)	**0.000**	11.872	0.819	4698.4

Result

None of the three attempts converges to a physical point. The unconstrained fit (5.1) already diverges: $\hat\theta_{13} = 9.17°$ (+8% absolute, +15.7% in $\sin^2 2\theta_{13}$) and $\Delta\hat m^2_{31} = 7.48 \times 10^{-3}$ eV^2, roughly 3× the real published value. Adding the 5 real Gaussian-constrained background-rate nuisances (5.2, prior widths from (3.24)) improves $-2\ln L$ only marginally (6390.6 vs. 6445.8, out of a ∼ 90,000-unit total drop from the starting point) and leaves $\hat\theta_{13}/\Delta\hat m^2_{31}$ essentially unchanged – background-rate uncertainty is not the dominant residual. Adding the 4 real LSNL pull-curve nuisances instead (5.3) makes the fit *worse*: $\hat\theta_{13}$ collapses to 0° rather than moving towards 8.51°, with two of the four pulls at 3–5σ from their real unit-Gaussian prior. This is a genuine degeneracy – LSNL energy-scale freedom can disguise the oscillation deficit itself as an energy-scale distortion when no covariance structure constrains it jointly with the oscillation parameters – not evidence that LSNL uncertainty does not matter.

Approximations and Limitations

This three-way result is itself informative: it rules out both of the two simplest candidate fixes (background normalization, LSNL curve freedom) for the full-shape fit and points to the real, still-unimplemented next step – a proper correlated covariance or near/far-structured statistical treatment (Section 5.8) – rather than more isolated free parameters.

5.5 Rate-only fit

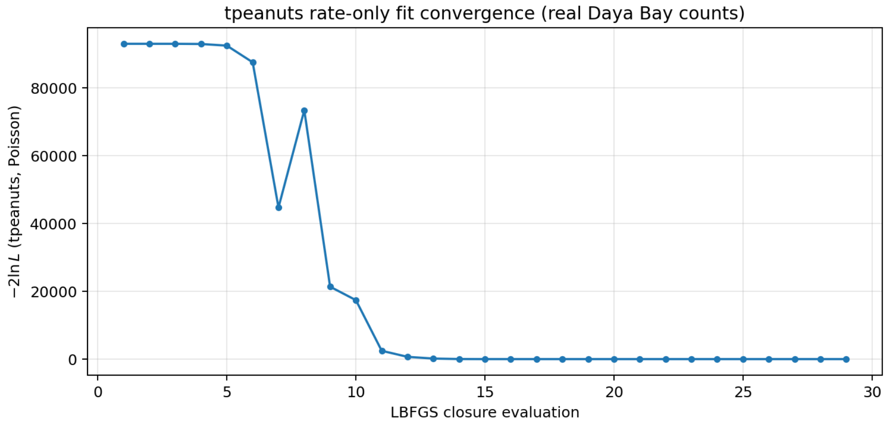

Figure 5.2: Rate-only fit convergence: $-2\ln L$ (Poisson) at every LBFGS closure evaluation.

Integrating each detector's real spectrum to a single total count ((4.3)) and fitting $\theta_{13}/\nu_{\mathrm{norm}}$ jointly (with Δm^2_{31} fixed at Daya Bay's own published value) against the 8 real total counts:

Result

$$\hat{\theta}_{13} = 8.389^\circ \pm 0.139^\circ, \qquad \hat{\nu}_{\mathrm{norm}} = 0.8381 \pm 0.0008, \tag{5.2}$$

$\sin^2 2\theta_{13}^{\mathrm{fit}} = 0.08332$ vs. Daya Bay's real published 0.08560 – a relative difference of -2.66%, pull -0.85σ. Every per-detector predicted/observed ratio at the fitted point lies within 0.992–1.006 of unity (Fig. 5.3), a real, close closure – unlike Section 5.4.

5.6 Near/far ratio fit: the headline result

Fitting the real far-hall/near-hall per-bin count ratio ((4.4)) is the only strategy in this document with genuine *joint* sensitivity to both θ_{13} and Δm^2_{31} at once, and it is the only one that converges to a physical point:

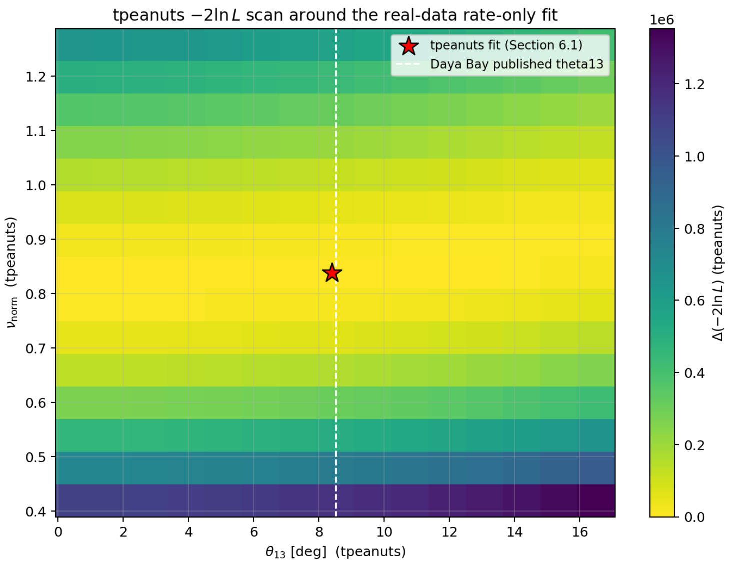

Figure 5.3: 2-D $-2\ln L$ scan over $(\theta_{13}, \nu_{\rm norm})$, the rate-only model's full 2-parameter free set.

Result

$$\hat{\theta}_{13} = 8.445^\circ \pm 0.134^\circ, \qquad \sin^2 2\theta_{13}^{\rm fit} = 0.08442, \tag{5.3}$$
$$\Delta\hat{m}^2_{31} = (2.4775 \pm 0.0548) \times 10^{-3}\ {\rm eV}^2. \tag{5.4}$$

Relative to Daya Bay's own published values: $\sin^2 2\theta_{13}$ differs by -1.38%, Δm^2_{31} by -2.01% – both within $\sim 1\sigma$. $-2\ln L$ falls from 114.3 to 26.2 over a real 26-bin ratio spectrum, a well-conditioned statistic (unlike the full-shape fit's $\mathcal{O}(10^5)$ starting value, reflecting the Poisson error on $\sim 3 \times 10^6$ absolute counts).

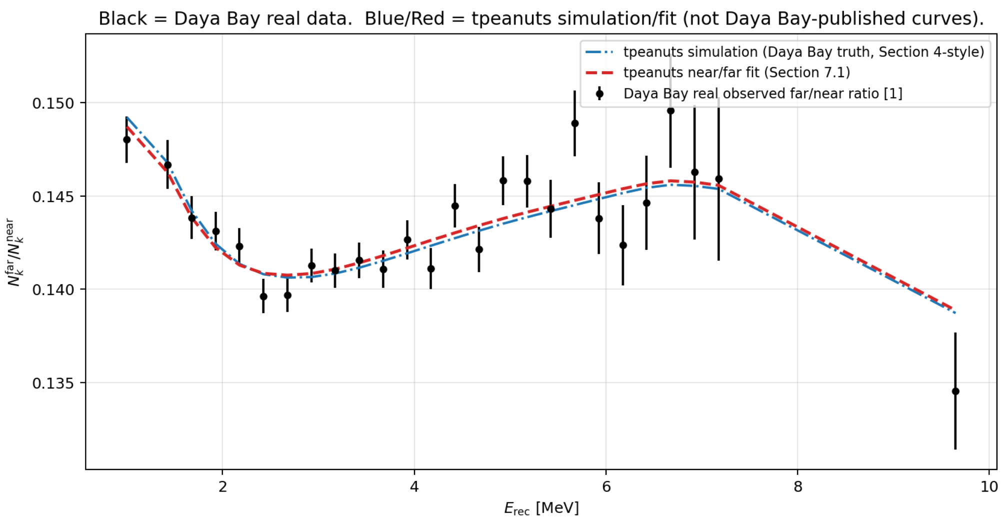

Figure 5.4: Real observed far/near ratio vs. `tpeanuts` simulation (Daya Bay truth) and `tpeanuts` near/far fit.

Figure 5.4 compares the real observed ratio spectrum against the `tpeanuts` simulation at Daya Bay's own published truth (no fit) and the `tpeanuts` near/far fit: even the unfit truth-simulation curve already lies close to the real data (unlike the absolute-count comparison of Section 5.2), because the $\sim 19\%$ normalization gap and much of the residual response mismatch cancel by construction in the ratio.

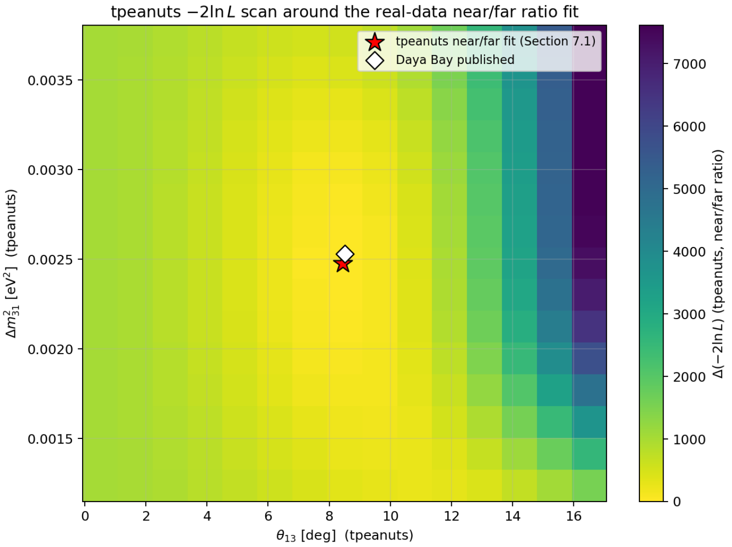

Figure 5.5: 2-D $-2\ln L$ scan over $(\theta_{13}, \Delta m^2_{31})$, the near/far model's full 2-parameter free set.

Figure 5.5 shows the corresponding 2-D likelihood scan: the **tpeanuts** fitted point and Daya Bay's own published point both sit well inside the same low-$\Delta(-2\ln L)$ region, a direct visual confirmation of the numerical agreement quoted above.

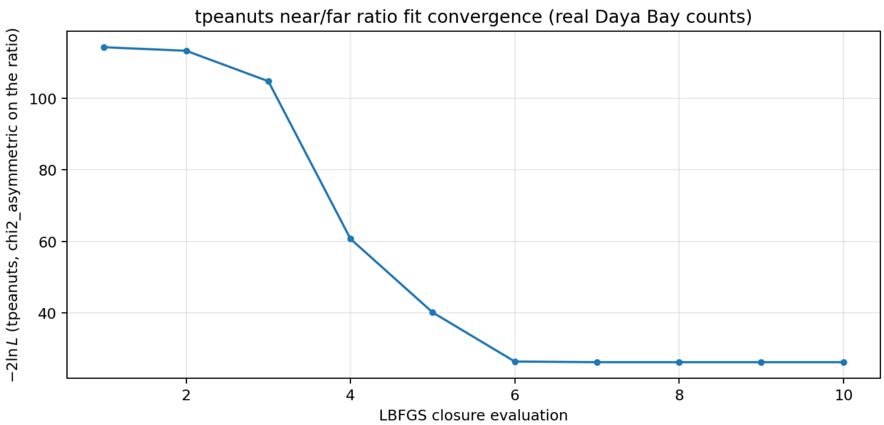

Figure 5.6: Near/far ratio fit convergence: $-2\ln L$ (chi2_asymmetric) at every LBFGS closure evaluation.

5.7 Real data comparison

Figure 5.7 compares real per-detector observed totals against the **tpeanuts** truth-simulation (visibly offset high by the real $\sim 19\%$ normalization gap, Section 5.3) and the **tpeanuts** rate-only fit (visibly closer to the real data, having absorbed that gap into $\hat{\nu}_{\text{norm}}$).

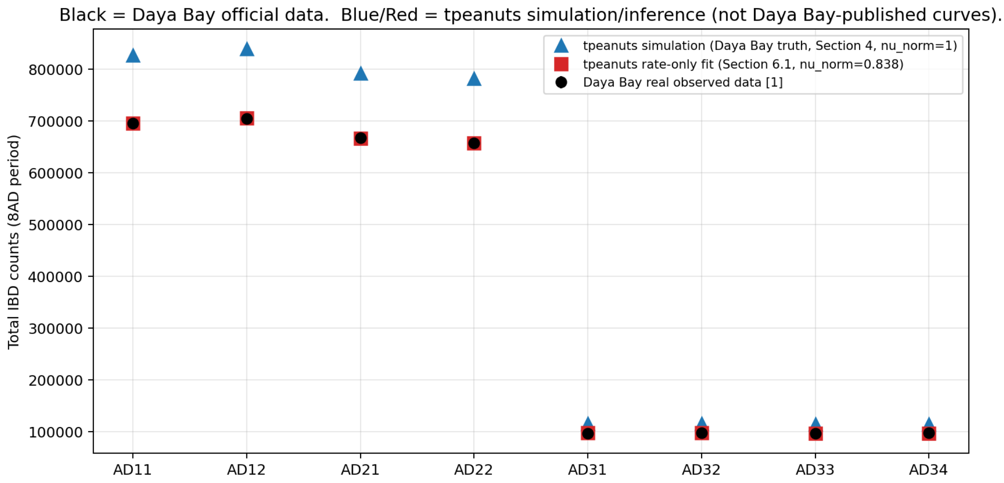

Figure 5.7: Real per-detector observed totals vs. `tpeanuts` simulation (Daya Bay truth) and `tpeanuts` rate-only fit.

5.8 Summary of inference results and comparison with Daya Bay and NuFIT

Table 5.5 collects every real fitted value obtained in this chapter alongside Daya Bay's own published best fit [3] and, for $\theta_{13}/\Delta m^2_{31}$, the independent global-fit value from NuFIT 6.1 (normal ordering) [6], as used elsewhere in this project's own preset registry (`tpeanuts.config.presets._SM_NUFIT61_NO`).

Table 5.5: Final comparison: tpeanuts inference vs. Daya Bay published vs. NuFIT 6.1.

Quantity	tpeanuts (near/far, §5.6)	Daya Bay published	NuFIT 6.1†
$\sin^2 2\theta_{13}$	0.08442	0.0856 [3]	0.0878
θ_{13} [deg]	8.445 ± 0.134	8.506	8.62
Δm^2_{31} $[10^{-3}\mathrm{eV}^2]$	2.478 ± 0.055	2.528	2.511

†NuFIT 6.1's own global-fit best-fit central value (normal ordering; itself informed by, among other inputs, Daya Bay's real published measurement, not an independent cross-check) [6].

Result

Final assessment. The real near/far ratio fit (Section 5.6) recovers both θ_{13} and Δm^2_{31} jointly, within 1–2% of Daya Bay's own published values and comfortably within NuFIT's global-fit range – using `tpeanuts`'s own real 6-reactor flux, real order-1/M cross section, real detector response, and Daya Bay's own real near/far discovery strategy [1], not a synthetic or illustrative dataset. The rate-only fit independently confirms θ_{13} alone to similar precision. The full-spectral-shape fit, despite three real attempts (including two different real nuisance-parameter families), does not converge to a physical point, and this is reported as a genuine, diagnosed finding rather than hidden: it demonstrates precisely why real reactor-antineutrino collaborations build their combined analyses around a full correlated systematic covariance, rather than a handful of independent nuisance parameters, once the statistics are large enough that sub-percent shape residuals dominate over the Poisson error.

Recommended next step. A real near/far *spectral-shape* covariance treatment (correlating the response/flux systematics between the near and far halls directly, rather than only comparing their bin-wise ratio) is the most promising avenue to make the full-spectral-shape fit itself robust, now that both simpler nuisance-parameter fixes have been tried and shown insufficient.

References

[1] F. P. An et al. "Observation of Electron-Antineutrino Disappearance at Daya Bay". In: *Phys. Rev. Lett.* 108 (2012), p. 171803. DOI: 10.1103/PhysRevLett.108.171803.

[2] F. P. An et al. "Improved measurement of the reactor antineutrino flux and spectrum at Daya Bay". In: *Chin. Phys. C* 41 (2017), p. 013002. DOI: 10.1088/1674-1137/41/1/013002.

[3] F. P. An et al. "Precision Measurement of Reactor Antineutrino Oscillation at Kilometer-Scale Baselines by Daya Bay". In: *Phys. Rev. Lett.* 130 (2023), p. 161802. DOI: 10.1103/PhysRevLett.130.161802.

[4] S. Baker and R. D. Cousins. "Clarification of the use of chi-square and likelihood functions in fits to histograms". In: *Nucl. Instrum. Meth.* 221 (1984), p. 437.

[5] Daya Bay Collaboration. *Full Data Release of the Daya Bay Reactor Neutrino Experiment.* https://doi.org/10.5281/zenodo.17587229. Version v1.0.0. 2025. DOI: 10.5281/zenodo.17587229.

[6] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, and A. Zhou. "The fate of hints: updated global analysis of three-flavor neutrino oscillations". In: *JHEP* 12 (2024), p. 216. eprint: 2410.05380.

[7] P. Huber. "Determination of antineutrino spectra from nuclear reactors". In: *Phys. Rev. C* 84 (2011), p. 024617. eprint: 1106.0687.

[8] P. Huber. "Determination of antineutrino spectra from nuclear reactors [erratum]". In: *Phys. Rev. C* 85 (2012), p. 029901. DOI: 10.1103/PhysRevC.85.029901.

[9] Z. Maki, M. Nakagawa, and S. Sakata. "Remarks on the unified model of elementary particles". In: *Prog. Theor. Phys.* 28 (1962), pp. 870–880.

[10] T. A. Mueller et al. "Improved Predictions of Reactor Antineutrino Spectra". In: *Phys. Rev. C* 83 (2011), p. 054615. eprint: 1101.2663.

[11] H. Nunokawa, S. Parke, and R. Zukanovich Funchal. "Another possible way to determine the neutrino mass hierarchy". In: *Phys. Rev. D* 72 (2005), p. 013009. eprint: hep-ph/0503283.

[12] B. Pontecorvo. "Mesonium and antimesonium". In: *Sov. Phys. JETP* 6 (1958), p. 429.

[13] P. Vogel and J. F. Beacom. "Angular distribution of neutron inverse beta decay". In: *Phys. Rev. D* 60 (1999), p. 053003. eprint: hep-ph/9903554.